

NETWORKS AND COMPLEXITY

Solution 16-7

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 16.7: Pitchfork normal form [Lvl. 3]

Consider a system $\dot{x} = f(x, p)$ with mirror anti-symmetry such that $f(x, p) = -f(-x, p)$. Show that there is a steady state at $x^* = 0$, then consider a bifurcation of this steady state that occurs at p^* . Follow the steps from Ex. 16.6 to derive the normal form of this bifurcation and analyze it.

Solution

As already hinted in the text we note substituting $x = 0$ into the symmetry relation yields

$$f(0, p) = -f(0, p) \quad (1)$$

which implies

$$f(0, p) = 0 \quad (\text{Stationarity condition}) \quad (2)$$

which proves that $x^* = 0$ is a steady state. Moreover the same argument can be repeated with every derivative that is of even order in x . For example differentiating the symmetry relation twice with respect to x and once with respect to p yields

$$f_{\text{xxp}}(x, p) = -f_{\text{xxp}}(-x, p) \quad (3)$$

such that

$$f_{\text{xxp}}(0, p) = -f_{\text{xxp}}(0, p) \quad (4)$$

which again implies

$$f_{\text{xxp}}(0, p) = 0. \quad (5)$$

We now focus on a bifurcation of the steady state at $x^* = 0$ and define the parameter value where the bifurcation occurs as p^* . Since this is a one dimensional system, the only way in which a bifurcation can occur is when the first derivative of the equation of motion becomes zero, so

$$f_x(x^*, p^*) = 0 \quad (\text{Bifurcation condition}) \quad (6)$$

We can now write

$$x = x^* + \delta \quad p = p^* + \rho \quad (7)$$

which defines ρ as the distance from the bifurcation and δ as the deviation from the steady state. For this bifurcation we could have just used x instead of δ as $x^* = 0$, but we will use δ anyway to maintain consistency with the previous derivations.

We can now write the following differential equation for the deviation:

$$\dot{\delta} = f(x^* + \delta, p^* + \rho) \quad (8)$$

We are interested in the behavior close to the bifurcation where δ and ρ are small. Hence, we Taylor expand and arrive at

$$\dot{\delta} = f + f_x \delta + f_p \rho + f_{\text{px}} \delta \rho + O(\delta^2) + O(\rho^2), \quad (9)$$

where we have again dropped the argument (x^*, p^*) . In the expansion, the first term $f(x^*, p^*)$ is zero due to the stationarity condition. The second term is zero because it is an even (0) order derivative with respect to x , however, the third term will in general be non-zero. Hence, we assume

$$f_{px}(x^*, p^*) \neq 0 \quad (\text{Genericity Condition 1}). \quad (10)$$

At this point we have

$$\dot{\delta} = f_{xp}\delta\rho + O(\delta^2) + O(\rho^2). \quad (11)$$

Among the further terms ($O(\delta^2) + O(\rho^2)$) we can ignore all the terms that contain at least one δ and at least one ρ as these are negligible in comparison to $f_{xp}\delta\rho$. This means we now only have to deal with the ‘pure’ terms that contain either derivatives with respect to δ or ρ but not both. Fortunately all terms that only contain derivatives with respect to ρ are forbidden by the symmetry condition. This leaves us with

$$\dot{\delta} = f_{xp}\delta\rho + O(\delta^2). \quad (12)$$

Expanding the next order of $O(\delta^2)$ gives us

$$\dot{\delta} = f_{xp}\delta\rho + \frac{1}{2}f_{xx}\delta^2 + O(\delta^3), \quad (13)$$

but $f_{xx} = 0$ due to symmetry, so we need to consider one more order to arrive at

$$\dot{\delta} = f_{xp}\delta\rho + \frac{1}{6}f_{xxx}\delta^3 + O(\delta^4). \quad (14)$$

In general

$$f_{xxx}(x^*, p^*) \neq 0 \quad (\text{Genericity condition 2}) \quad (15)$$

Therefore we can now argue that all terms in $O(\delta^4)$ are negligible in comparison to $f_{xxx}\delta^3/6$, leaving us with

$$\dot{\delta} = f_{xp}\delta\rho + \frac{1}{6}f_{xxx}\delta^3. \quad (16)$$

To hide away the unknown constants we now define

$$p = f_{xp}(x^*, p^*)\rho \quad (17)$$

$$a = f_{xxx}(x^*, p^*)/6 \quad (18)$$

which allows us to write the desired normal form as

$$\dot{x} = px + ax^3. \quad (19)$$

Here I’ve switched back to x to make the dynamical system look more harmless. Normally this would require defining a ‘new’ x , but not in this case as $x = \delta$ anyway due to $x^* = 0$.

Now that we have the normal form we only need to analyze it. To find the steady states we set the equation of motion to zero which gives us

$$0 = px + ax^3. \quad (20)$$

We already know that there is a steady state at $x_1^* = 0$, and dividing this steady state out of the equation leaves us with

$$0 = p + ax^2. \quad (21)$$

which we can solve for

$$x_{2,3}^* = \pm \sqrt{-\frac{p}{a}}. \quad (22)$$

So there are two additional branches of steady states that only exist for $-p/a > 0$.

Next we consider the stability of the steady states. In the one-dimensional system a steady state is stable if the derivative of the equation of motion with respect to x is negative in the steady state, so when

$$p + 3ax^2 < 0 \quad (23)$$

Therefore, the trivial steady state $x_1^* = 0$ is stable if $p < 0$ and unstable otherwise. Substituting the other two steady states into the condition is a bit tricky as we have to be careful with the sign of the second term. However we know that the steady states only exist when $-p/a > 0$. In this case substituting the steady states makes the square and the square root cancel and we get the positive term $-p/a$ back (including the minus). So, substituting the steady states into the stability condition yields

$$p + 3a(-p/a) = -2p < 0. \quad (24)$$

Therefore the nontrivial steady states $x_{2,3}^*$ are stable when they occur at $p > 0$.

Considering again the condition for the existence of the steady states $-p/a > 0$ we note that we can meet the condition if either $a < 0$ or $p < 0$ but not both. Hence we distinguish two cases. If $a < 0$ then the nontrivial steady states exist when $p > 0$. In this case the nontrivial steady states are stable (supercritical case). Conversely, if $a > 0$ then the nontrivial steady states exist if $p < 0$ and they are unstable (subcritical case).

We can now draw the bifurcation diagrams from the chapter, which depict the cases $a = 1$ and $a = -1$.